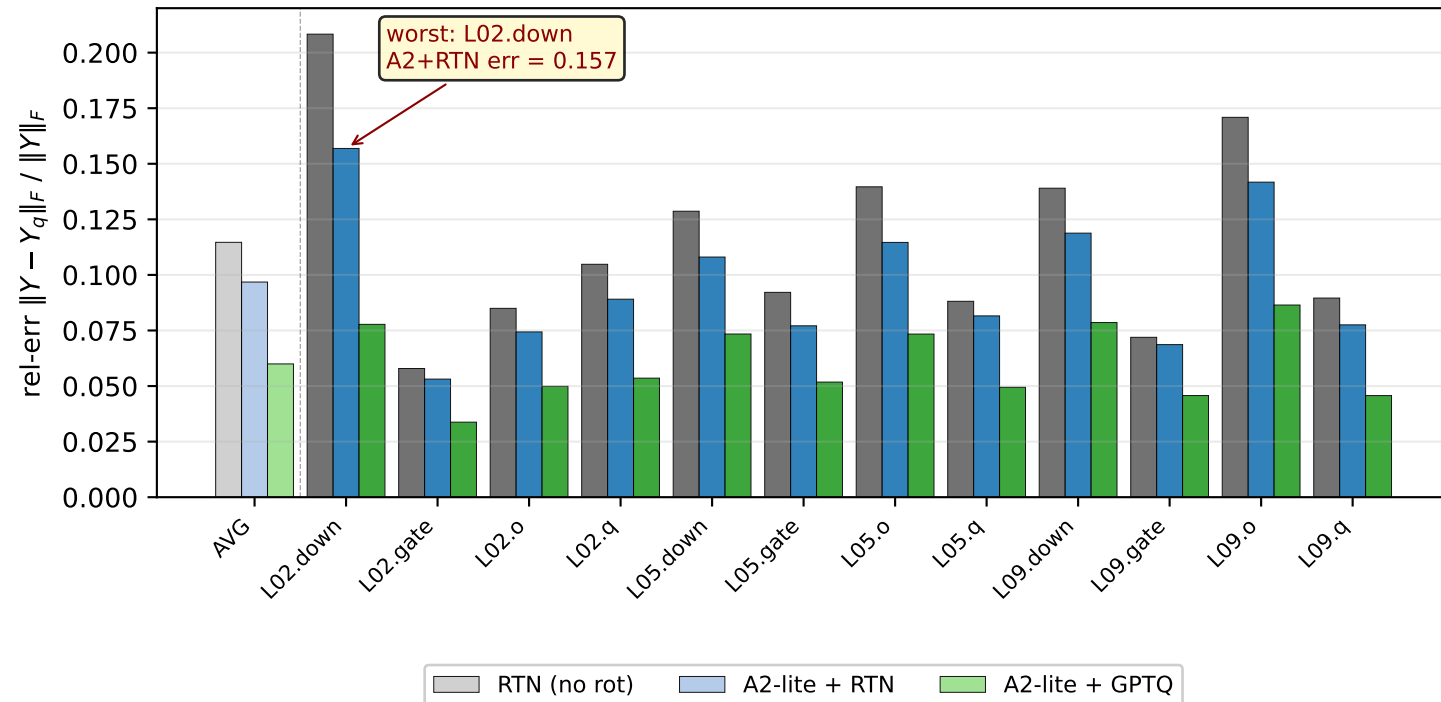
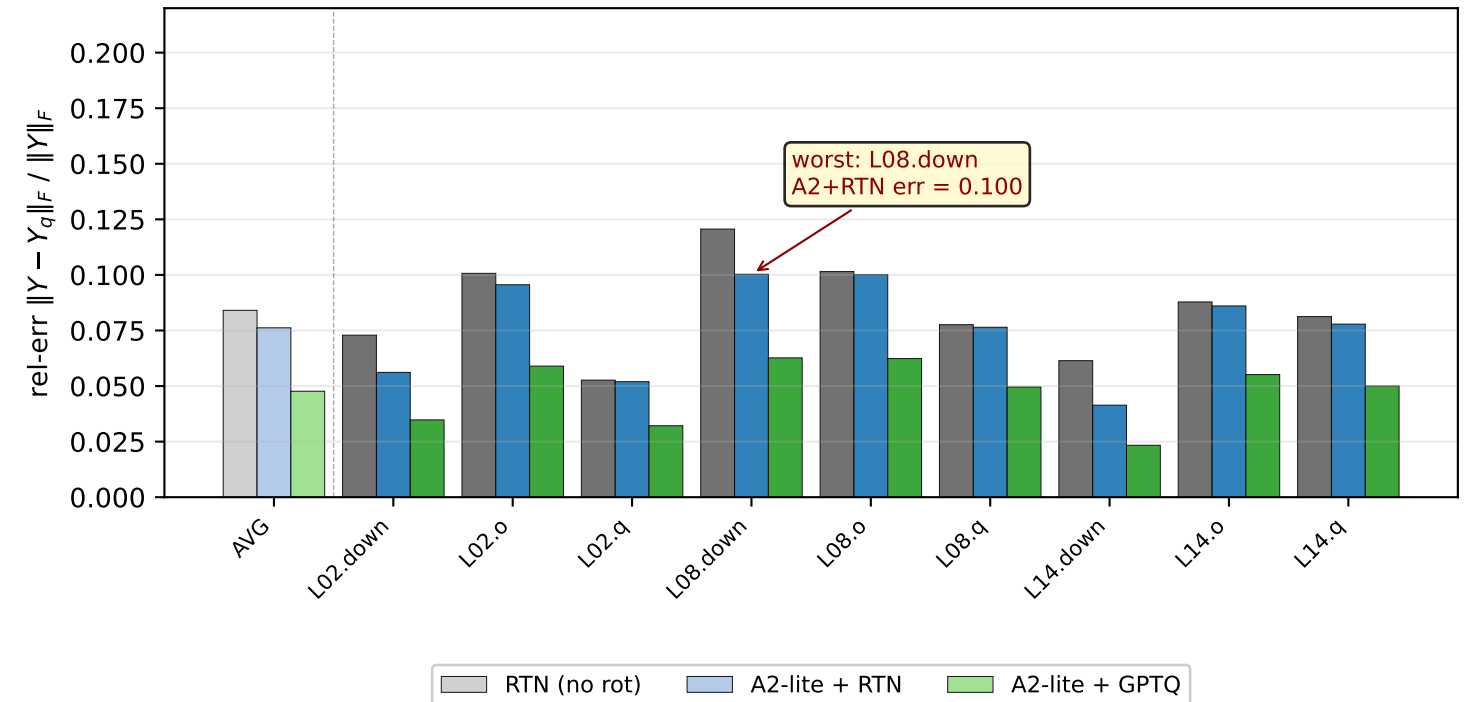


W4 weight-quant error per component: RTN vs A2-lite + RTN vs A2-lite + GPTQ (weight-only error; activation quant excluded — isolates structural effect of each step)

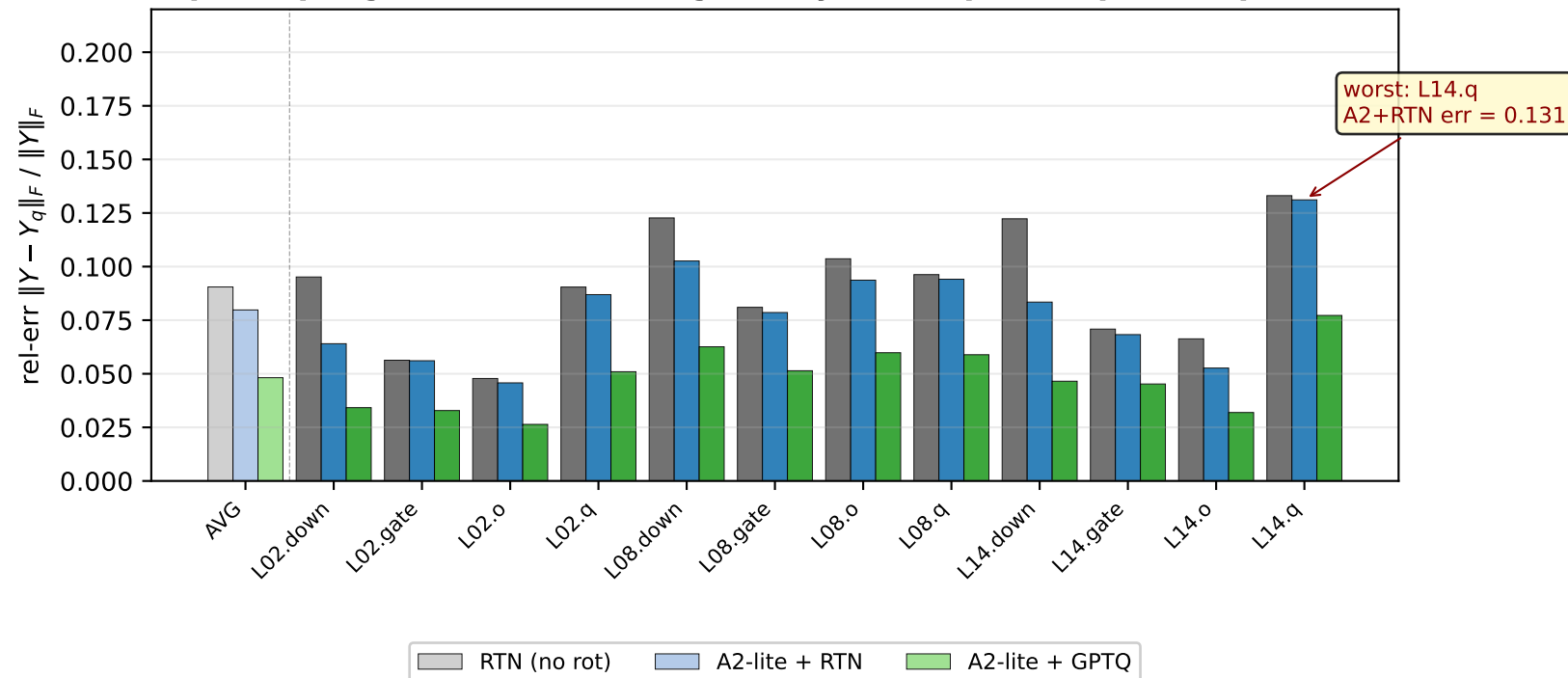
gr00t-N1.5 LLM (eagle) — W4 weight-only rel-output-err per component



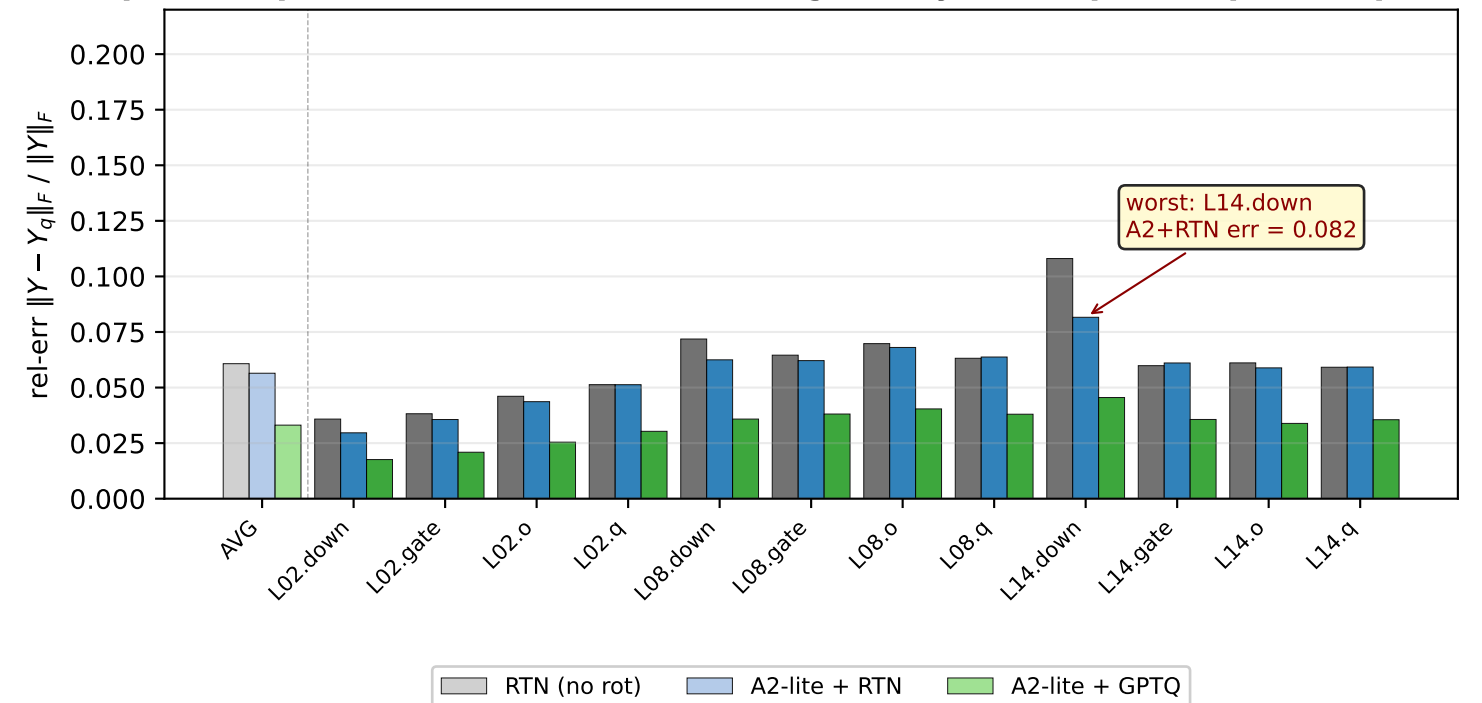
gr00t-N1.5 DiT — W4 weight-only rel-output-err per component



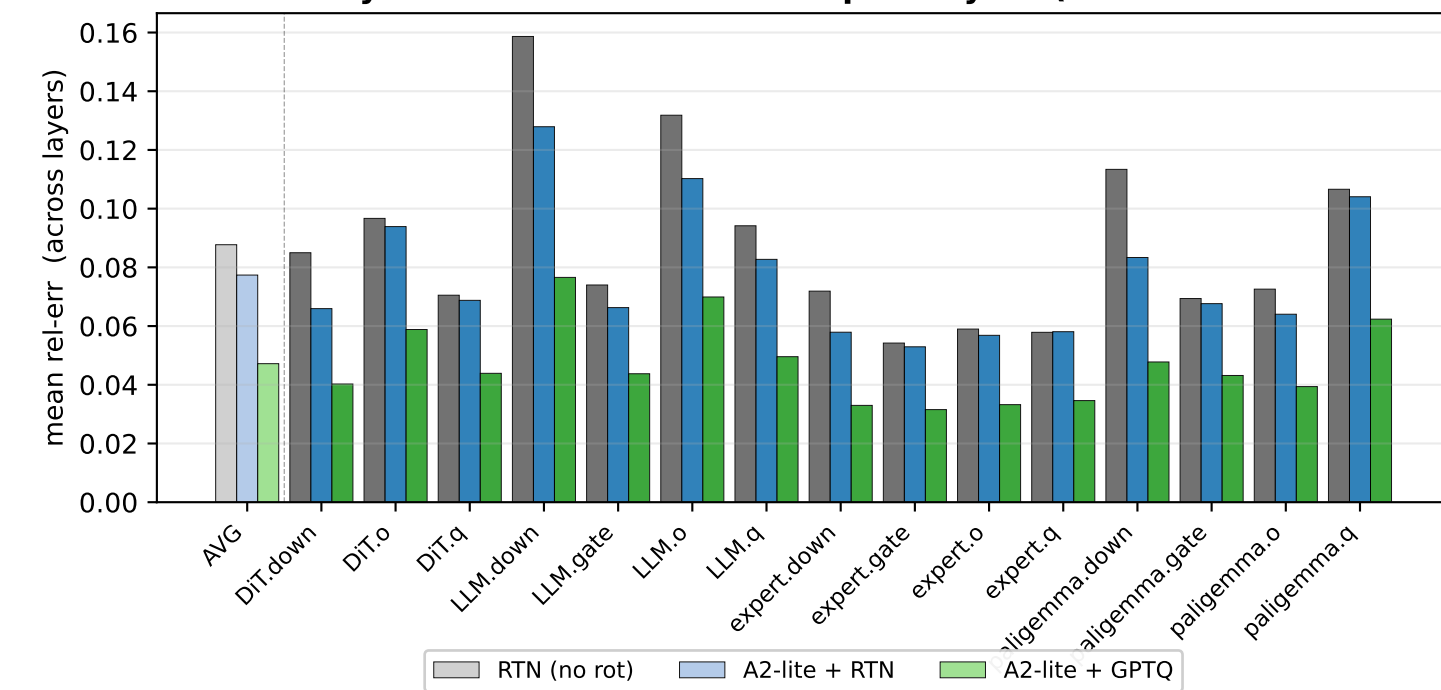
pi0.5 paligemma — W4 weight-only rel-output-err per component



pi0.5 expert / action head — W4 weight-only rel-output-err per component



Per-kind summary: mean rel-err across sampled layers (AVG = mean over all kinds)



**Isolated GPTQ contribution after rotation:
GPTQ over RTN reduces W4 weight-only rel-err ~40-50%**

